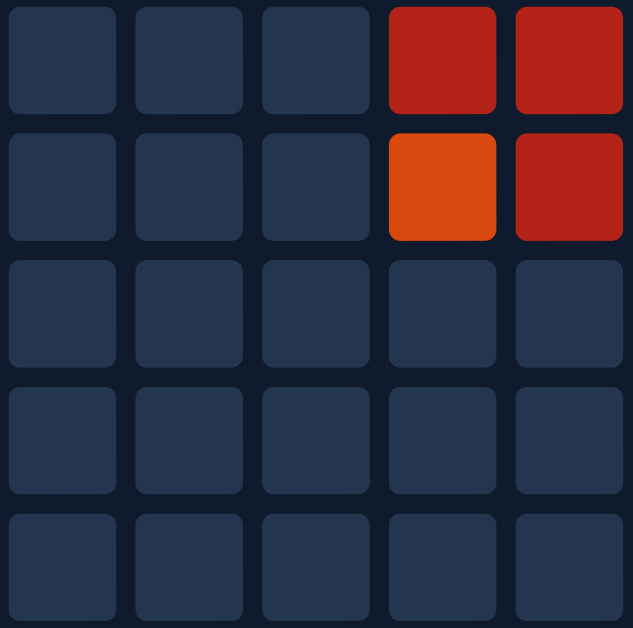
CLOUD SECURITY ASSESSMENT

Acme Corp (sample data)

Configuration risks, sensitive data exposure and suspicious activity

6 November 2026



Based on configuration findings, data inventory and authentication logs

OVERVIEW

Executive summary

The account has a small number of serious, connected problems rather than widespread weakness. The most urgent is a public storage bucket holding 48,200 customer records that was accessed from a new country on 29 October. A payments database is also reachable from the internet, and the root account has no multi-factor authentication.

40

findings across 20 assets

15

critical or high severity

3 of 8

sensitive data stores exposed

3

suspicious access events

What to do first

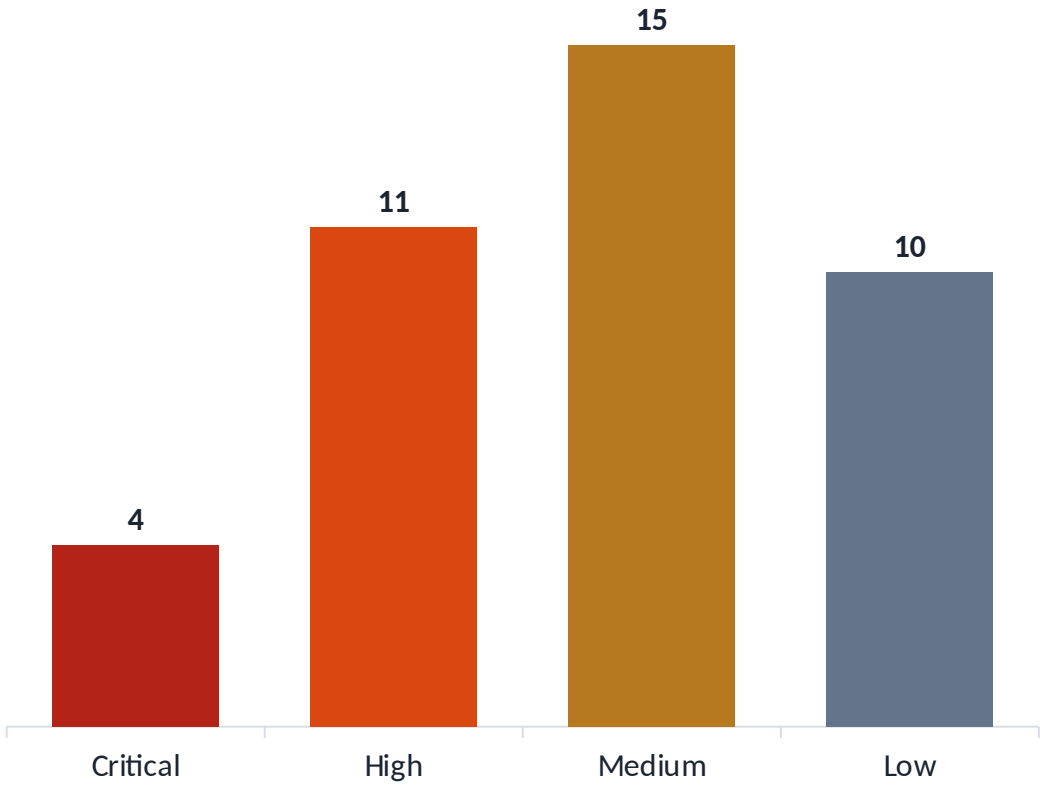
1. Close public access to customer and payment data first
2. Treat the 29 October access to customer exports as a possible incident
3. Secure the root account and restore audit logging in every region

SECURITY POSTURE

15 of 40 findings are critical or high severity

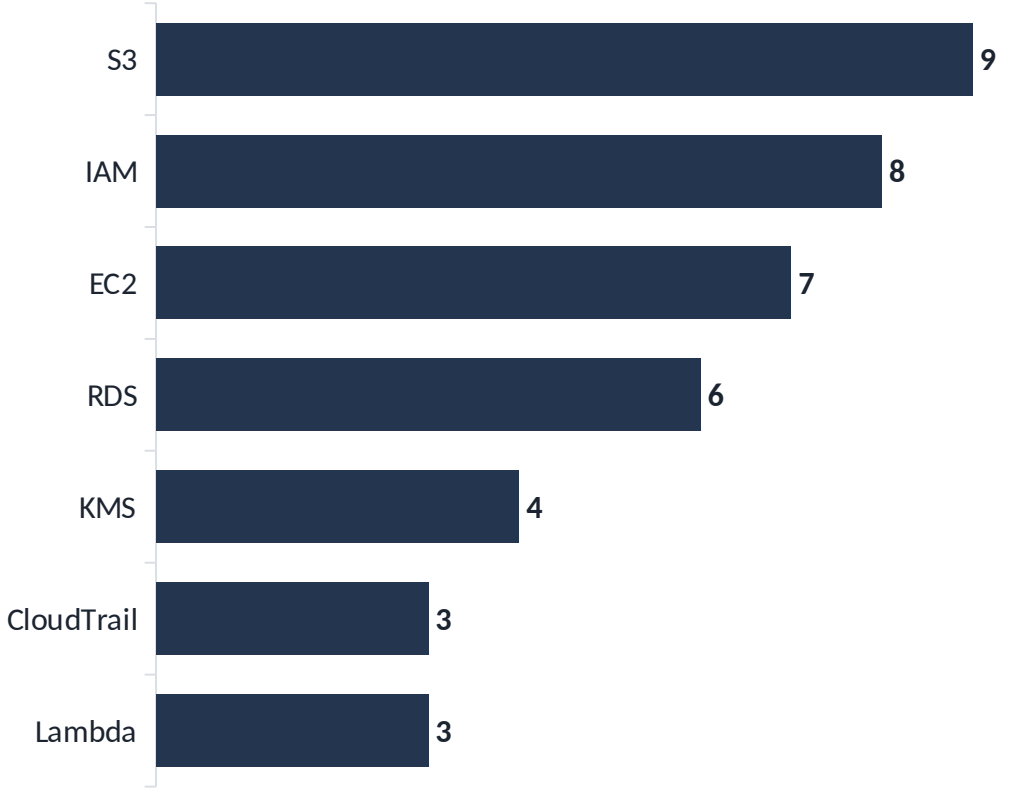
Findings by severity

38% need action within 60 days



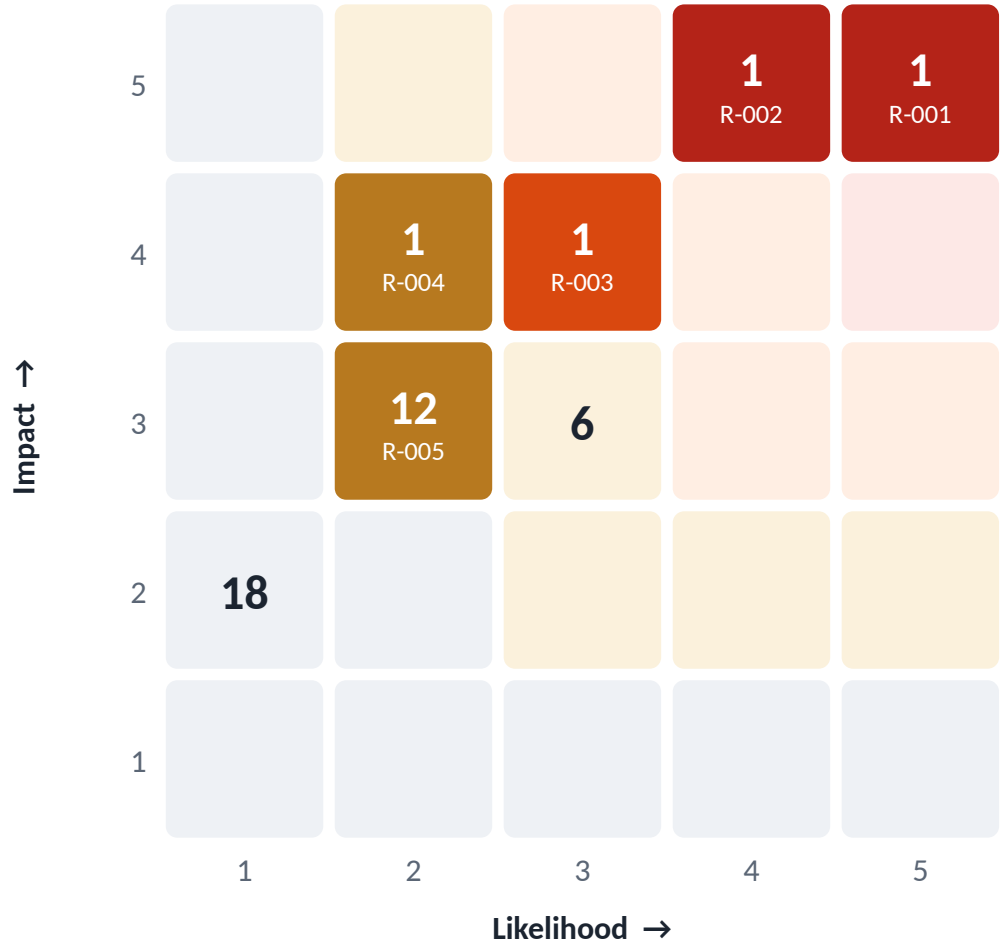
Findings by cloud service

Where the configuration problems are concentrated



RISK HEATMAP

Most findings are low risk; a few need action now



- **Critical zone** · Act now
- **High zone** · Plan this month
- **Medium zone** · Schedule
- **Low zone** · Monitor

Numbers show how many findings sit in each cell. Solid cells contain a top risk.

Top risks on the map

R-001	Public storage bucket holding customer personal data
R-002	Payments database reachable from the internet
R-003	Root account without multi-factor authentication
R-004	Unencrypted backups of the HR database
R-005	Audit logging disabled in one region

SENSITIVE DATA EXPOSURE

3 of 8 sensitive data stores are exposed

Records at risk

173,200

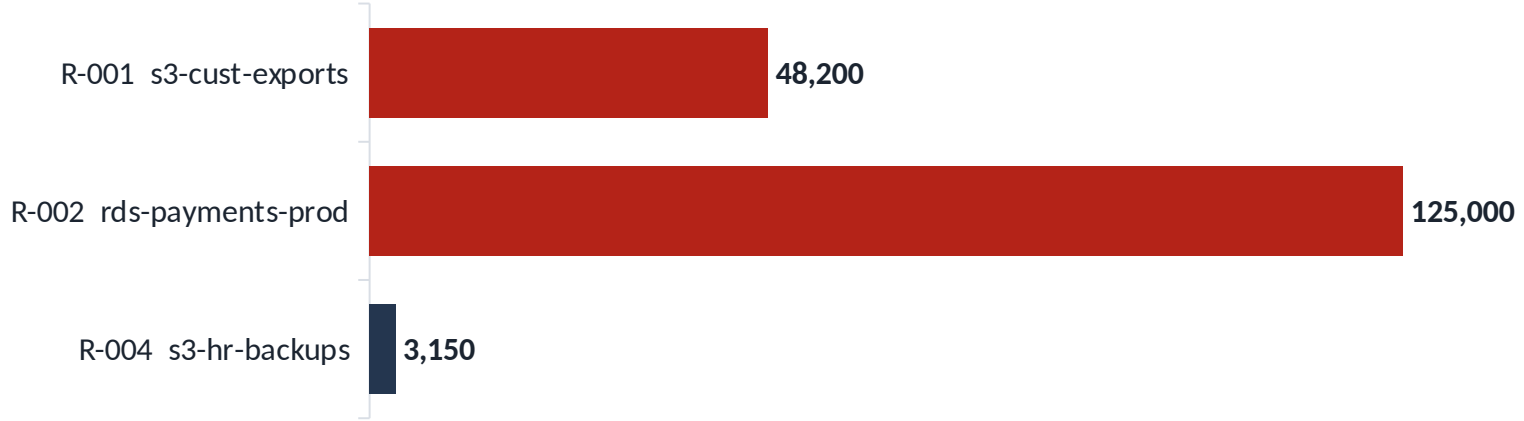
personal or financial records held in publicly exposed stores among the top risks

Exposure combined with sensitive data doubles a risk's score; suspicious access raises it further.

Risk	Asset	Data	Records	Exposure	Suspicious access
R-001	s3-cust-exports	PII	48,200	Public	Yes
R-002	rds-payments-prod	financial	125,000	Public	No
R-004	s3-hr-backups	PII	3,150	Internal	No

Records held per data store

red = publicly exposed



SUSPICIOUS ACTIVITY

3 suspicious access events were detected

Login from new country

User	**svc-export**
Asset	**s3-cust-exports**
Country	**RU**
When	**29 Oct 2026, 02:41 UTC**

First login from this country in 90 days

Rule: First login from this country for the user in the last 90 days

Linked to R-001

Impossible travel

User	**j.mwangi**
Asset	**console**
Country	**BR**
When	**27 Oct 2026, 09:15 UTC**

Login from Nairobi and São Paulo 40 minutes apart

Rule: Two logins too far apart to travel between in the time given (over 900 km/h)

Off-hours admin login

User	**root**
Asset	**iam-root**
Country	**KE**
When	**25 Oct 2026, 00:12 UTC**

Root login at 03:12 local time

Rule: Admin or root login outside 08:00-18:00 local time

Linked to R-003

TOP RISK 1 OF 3 · S3

Public storage bucket holding customer personal data

CRITICAL

Risk score

60

Severity **critical**	**10**
Exposure **public**	**× 2**
Data **PII**	**× 2**
Activity **suspicious**	**× 1.5**
Score	**= 60**

Asset: s3-cust-exports

Evidence

CONFIGURATION FINDING F-012

Bucket policy allows public read

DATA INVENTORY s3-cust-exports

48,200 customer PII records

AUTHENTICATION LOGS A-0417

Access from new country (RU) on 2026-10-29

Business impact

Possible exposure of customer personal data, with regulatory and reputational consequences.

Recommended fix

Block public access, rotate access keys, review access logs since 2026-10-01.

Complete within 0–30 days

TOP RISK 2 OF 3 · RDS

Payments database reachable from the internet

CRITICAL

Risk score

40

Severity **critical**	**10**
Exposure **public**	**× 2**
Data **financial**	**× 2**
Activity **normal**	**× 1**
Score	**= 40**

Asset: rds-payments-prod

Evidence

CONFIGURATION FINDING F-021

Security group allows 0.0.0.0/0 on port 5432

DATA INVENTORY rds-payments-prod

125,000 payment records

Business impact

Financial records exposed to internet-wide attack attempts.

Recommended fix

Restrict inbound access to the application subnet and enable IAM database authentication.

Complete within 0–30 days

Root account without multi-factor authentication

CRITICAL

Risk score

18

Severity **critical**	**10**
Exposure **internal**	**× 1**
Data **internal**	**× 1.2**
Activity **suspicious**	**× 1.5**
Score	**= 18**

Asset: iam-root

Evidence

CONFIGURATION FINDING F-003

MFA not enabled on root account

AUTHENTICATION LOGS A-0388

Root login at 03:12 local time

Business impact

A stolen password would give full control of the cloud account.

Recommended fix

Enable hardware MFA on root and stop using root for daily tasks.

Complete within 0–30 days

CORRELATED RISK

Three signals, one asset: why R-001 ranks first

1

Misconfiguration

CONFIGURATION FINDING F-012

Bucket policy allows public read

2

Sensitive data

DATA INVENTORY S3-CUST-EXPORTS

48,200 customer PII records

3

Suspicious access

AUTHENTICATION LOGS A-0417

Access from new country (RU) on 2026-10-29

Combined risk score

60

Rank 1 of 5 top risks · s3-cust-exports

No single signal here is unusual on its own. Together they show a public bucket, holding customer personal data, being accessed by a service account from a country it has never used. That combination is why this risk ranks first.

NEXT 90 DAYS

Remediation roadmap

0–30 days

Immediate

R-001 · CRITICAL

Public storage bucket holding customer personal data

Block public access, rotate access keys, review access logs since 2026-10-01.

R-002 · CRITICAL

Payments database reachable from the internet

Restrict inbound access to the application subnet and enable IAM database authentication.

R-003 · CRITICAL

Root account without multi-factor authentication

Enable hardware MFA on root and stop using root for daily tasks.

31–60 days

Next

R-004 · HIGH

Unencrypted backups of the HR database

Enable default encryption with a managed KMS key and re-encrypt existing objects.

61–90 days

Planned

R-005 · MEDIUM

Audit logging disabled in one region

Enable a multi-region trail with log file validation.

METHODOLOGY

How this assessment was produced

Risk scoring

Risk = severity × exposure × data sensitivity × activity

Factor	Values
Severity	critical 10 high 7 medium 4 low 1
Exposure	public 2 internal 1
Data sensitivity	PII 2 financial 2 health 2 internal 1.2 public 1
Activity	anomalous 1.5 normal 1

Likelihood: public = 4, internal = 2; +1 if anomalous access; max 5

Scoring model version 1.0 · data generated 6 November 2026

Data sources

Configuration findings
Cloud resources checked against security best practice

Data inventory
Which assets hold personal, financial or health data

Authentication logs
Who signed in, from where and when

Anomaly rules

Login from new country
First login from this country for the user in the last 90 days

Impossible travel
Two logins too far apart to travel between in the time given (over 900 km/h)

Off-hours admin login
Admin or root login outside 08:00-18:00 local time